

## 9.2 The Power Method

*The eigenvalues of a square matrix can, in theory, be found by solving the characteristic equation. However, this procedure has so many computational difficulties that it is almost never used in applications. In this section we discuss an algorithm that can be used to approximate the eigenvalue with greatest absolute value and a corresponding eigenvector. This particular eigenvalue and its corresponding eigenvectors are important because they arise naturally in many iterative processes. The methods studied here have recently been used to create Internet search engines such as Google.*

### The Power Method

When a vector  $\mathbf{x}_0$  in  $R^n$  is multiplied repeatedly by an  $n \times n$  matrix  $A$ , it produces the **power sequence generated by  $A$** :

$$\mathbf{x}_0, \quad A\mathbf{x}_0, \quad A^2\mathbf{x}_0, \quad \dots, \quad A^k\mathbf{x}_0, \quad \dots$$

#### Definition 1

If the distinct eigenvalues of  $A$  are  $\lambda_1, \lambda_2, \dots, \lambda_k$ , and if  $|\lambda_1|$  is larger than  $|\lambda_2|, \dots, |\lambda_k|$ , then  $\lambda_1$  is called a **dominant eigenvalue** of  $A$ . Any eigenvector corresponding to a dominant eigenvalue is called a **dominant eigenvector** of  $A$ .

#### EXAMPLE 1 | Dominant Eigenvalues

Dominant eigenvalues do not always exist:

- If  $\lambda_1 = -4$ ,  $\lambda_2 = -2$ ,  $\lambda_3 = 1$ ,  $\lambda_4 = 3$ , then  $\lambda_1 = -4$  is dominant since  $|\lambda_1| = 4$  exceeds all other absolute values.
- If  $\lambda_1 = 7$ ,  $\lambda_2 = -7$ ,  $\lambda_3 = -2$ ,  $\lambda_4 = 5$ , then  $|\lambda_1| = |\lambda_2| = 7$ , so *no* dominant eigenvalue exists.

We limit discussion to  $n \times n$  matrices with  $n$  linearly independent eigenvectors (symmetric matrices, for example).

### Theorem 9.2.1

Let  $A$  be a symmetric  $n \times n$  matrix with a positive<sup>a</sup> dominant eigenvalue  $\lambda$ . If  $\mathbf{x}_0$  is a unit vector in  $R^n$  not orthogonal to the eigenspace of  $\lambda$ , then the normalized power sequence

$$\mathbf{x}_0, \quad \mathbf{x}_1 = \frac{A\mathbf{x}_0}{\|A\mathbf{x}_0\|}, \quad \mathbf{x}_2 = \frac{A\mathbf{x}_1}{\|A\mathbf{x}_1\|}, \quad \dots, \quad \mathbf{x}_k = \frac{A\mathbf{x}_{k-1}}{\|A\mathbf{x}_{k-1}\|}, \quad \dots \quad (1)$$

converges to a unit dominant eigenvector, and the sequence

$$A\mathbf{x}_0 \cdot \mathbf{x}_0, \quad A\mathbf{x}_1 \cdot \mathbf{x}_1, \quad \dots, \quad A\mathbf{x}_k \cdot \mathbf{x}_k, \quad \dots \quad (2)$$

converges to the dominant eigenvalue  $\lambda$ .

<sup>a</sup>If the dominant eigenvalue is not positive, sequence (2) still converges to  $\lambda$ , but (1) may not converge to a specific dominant eigenvector due to alternation.

*Remark.* Sequence (1) can also be expressed as

$$\mathbf{x}_k = \frac{A^k \mathbf{x}_0}{\|A^k \mathbf{x}_0\|}, \quad \dots \quad (3)$$

expressing each iterate directly from the starting vector  $\mathbf{x}_0$ .

**Geometric plausibility.** Split  $\mathbf{x}_0 = \mathbf{v}_0 + \mathbf{w}_0$ , where  $\mathbf{v}_0, \mathbf{w}_0$  are orthogonal projections on the eigenspaces of  $\lambda_1$  and  $\lambda_2$  ( $\lambda_1 > \lambda_2 > 0$ ). Then  $A\mathbf{x}_0 = \lambda_1 \mathbf{v}_0 + \lambda_2 \mathbf{w}_0$ . Since  $\lambda_1 > \lambda_2$ , repeated multiplication by  $A$  amplifies the  $\mathbf{v}_0$  component more than the  $\mathbf{w}_0$  component, pulling iterates toward the eigenspace of  $\lambda_1$ . Normalizing keeps all iterates on the unit circle.

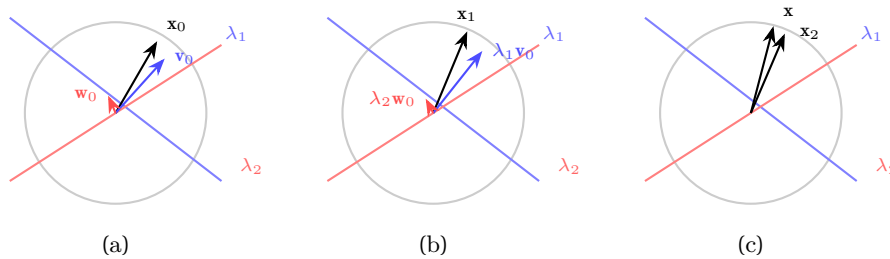


FIGURE 9.2.1 (a) Initial  $\mathbf{x}_0$ ; (b) after one multiplication; (c) iterates converge toward eigenspace of  $\lambda_1$

## The Power Method with Euclidean Scaling

Theorem 9.2.1 gives the following algorithm, called the **power method with Euclidean scaling**:

### The Power Method with Euclidean Scaling

**Step 0.** Choose an arbitrary nonzero vector; normalize it (if needed) to a unit vector  $\mathbf{x}_0$ .

**Step 1.** Compute  $A\mathbf{x}_0$  and normalize to get  $\mathbf{x}_1$ . Compute  $A\mathbf{x}_1 \cdot \mathbf{x}_1$  (first eigenvalue approximation).

**Step  $k$ .** Compute  $A\mathbf{x}_{k-1}$  and normalize to get  $\mathbf{x}_k$ . Compute  $A\mathbf{x}_k \cdot \mathbf{x}_k$  ( $k$ th eigenvalue approximation).

Continuing generates better and better approximations to  $\lambda$  and a unit dominant eigenvector.

## EXAMPLE 2 | The Power Method with Euclidean Scaling

Apply the power method with Euclidean scaling to  $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$ ,  $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ . Stop at  $\mathbf{x}_5$ .

**SOLUTION:** The eigenvalues of  $A$  are  $\lambda = 1$  and  $\lambda = 5$ ; the dominant eigenspace is  $x_1 = x_2$ , with unit dominant eigenvector  $\mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \approx \begin{bmatrix} 0.707 \dots \\ 0.707 \dots \end{bmatrix}$ .

| $k$ | $A\mathbf{x}_{k-1}$                                        | $\mathbf{x}_k$                                                                                                      | $\lambda^{(k)} = A\mathbf{x}_k \cdot \mathbf{x}_k$ |
|-----|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| 1   | $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$                     | $\frac{1}{\sqrt{13}} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \approx \begin{bmatrix} 0.83205 \\ 0.55470 \end{bmatrix}$ | $\approx 4.84615$                                  |
| 2   | $\approx \begin{bmatrix} 3.60555 \\ 3.32820 \end{bmatrix}$ | $\approx \begin{bmatrix} 0.73480 \\ 0.67828 \end{bmatrix}$                                                          | $\approx 4.99361$                                  |
| 3   | $\approx \begin{bmatrix} 3.56097 \\ 3.50445 \end{bmatrix}$ | $\approx \begin{bmatrix} 0.71274 \\ 0.70143 \end{bmatrix}$                                                          | $\approx 4.99974$                                  |
| 4   | $\approx \begin{bmatrix} 3.54108 \\ 3.52976 \end{bmatrix}$ | $\approx \begin{bmatrix} 0.70824 \\ 0.70597 \end{bmatrix}$                                                          | $\approx 4.99999$                                  |
| 5   | $\approx \begin{bmatrix} 3.53666 \\ 3.53440 \end{bmatrix}$ | $\approx \begin{bmatrix} 0.70733 \\ 0.70688 \end{bmatrix}$                                                          | $\approx 5.00000$                                  |

Thus  $\lambda^{(5)} \approx 5.00000$  (five-decimal-place accuracy) and  $\mathbf{x}_5$  approximates the dominant unit eigenvector  $\mathbf{v}_1$  to three decimal places.

## The Power Method with Maximum Entry Scaling

An alternative normalizes each iterate to make its maximum absolute entry equal to 1. Denote  $\max(\mathbf{x}) = \text{maximum absolute value of entries of } \mathbf{x}$ .

### Theorem 9.2.2

Let  $A$  be a symmetric  $n \times n$  matrix with a positive dominant eigenvalue  $\lambda$ . If  $\mathbf{x}_0$  is a nonzero vector in  $R^n$  not orthogonal to the eigenspace of  $\lambda$ , then the sequence

$$\mathbf{x}_0, \quad \mathbf{x}_1 = \frac{A\mathbf{x}_0}{\max(A\mathbf{x}_0)}, \quad \mathbf{x}_2 = \frac{A\mathbf{x}_1}{\max(A\mathbf{x}_1)}, \quad \dots, \quad \mathbf{x}_k = \frac{A\mathbf{x}_{k-1}}{\max(A\mathbf{x}_{k-1})}, \quad \dots \quad (8)$$

converges to an eigenvector corresponding to  $\lambda$ , and the sequence of **Rayleigh quotients**

$$\frac{A\mathbf{x}_0 \cdot \mathbf{x}_0}{\mathbf{x}_0 \cdot \mathbf{x}_0}, \quad \frac{A\mathbf{x}_1 \cdot \mathbf{x}_1}{\mathbf{x}_1 \cdot \mathbf{x}_1}, \quad \dots, \quad \frac{A\mathbf{x}_k \cdot \mathbf{x}_k}{\mathbf{x}_k \cdot \mathbf{x}_k}, \quad \dots \quad (9)$$

converges to  $\lambda$ .

Each term in (9) is the **Rayleigh quotient**

$$\frac{A\mathbf{x} \cdot \mathbf{x}}{\mathbf{x} \cdot \mathbf{x}} \quad (11)$$

When  $\mathbf{x}$  is a dominant eigenvector, the Rayleigh quotient equals  $\lambda$ .

### The Power Method with Maximum Entry Scaling

**Step 0.** Choose an arbitrary nonzero vector  $\mathbf{x}_0$ .

**Step  $k$ .** Compute  $A\mathbf{x}_{k-1}$ , scale by  $1/\max(A\mathbf{x}_{k-1})$  to get  $\mathbf{x}_k$ . Compute the Rayleigh quotient of  $\mathbf{x}_k$  as the  $k$ th eigenvalue approximation.

Continue until convergence.

### EXAMPLE 3 | Example 2 Revisited Using Maximum Entry Scaling

Apply the power method with maximum entry scaling to  $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$ ,  $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ .

**SOLUTION:**

| $k$ | $A\mathbf{x}_{k-1}$                                        | $\mathbf{x}_k$                                                                                              | $\lambda^{(k)} = \frac{A\mathbf{x}_k \cdot \mathbf{x}_k}{\mathbf{x}_k \cdot \mathbf{x}_k}$ |
|-----|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1   | $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$                     | $\frac{1}{3} \begin{bmatrix} 3 \\ 2 \end{bmatrix} \approx \begin{bmatrix} 1.00000 \\ 0.66667 \end{bmatrix}$ | $\approx 4.84615$                                                                          |
| 2   | $\approx \begin{bmatrix} 4.33333 \\ 4.00000 \end{bmatrix}$ | $\approx \begin{bmatrix} 1.00000 \\ 0.92308 \end{bmatrix}$                                                  | $\approx 4.99361$                                                                          |
| 3   | $\approx \begin{bmatrix} 4.84615 \\ 4.76923 \end{bmatrix}$ | $\approx \begin{bmatrix} 1.00000 \\ 0.98413 \end{bmatrix}$                                                  | $\approx 4.99974$                                                                          |
| 4   | $\approx \begin{bmatrix} 4.96825 \\ 4.95238 \end{bmatrix}$ | $\approx \begin{bmatrix} 1.00000 \\ 0.99681 \end{bmatrix}$                                                  | $\approx 4.99999$                                                                          |
| 5   | $\approx \begin{bmatrix} 4.99361 \\ 4.99042 \end{bmatrix}$ | $\approx \begin{bmatrix} 1.00000 \\ 0.99936 \end{bmatrix}$                                                  | $\approx 5.00000$                                                                          |

The Rayleigh quotients  $\lambda^{(k)}$  agree with those from Euclidean scaling. Maximum entry scaling produces an eigenvector approaching  $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$  (set  $t = 1$  in the parametric form (6)).

### Rate of Convergence

If  $|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_k|$ , the rate at which Rayleigh quotients converge to  $\lambda_1$  depends on the ratio  $|\lambda_1|/|\lambda_2|$ . The larger the ratio, the faster the convergence. In Example 3,  $|\lambda_1|/|\lambda_2| = 5/1 = 5$ , which is why convergence is rapid.

### Stopping Procedures

If  $\lambda$  is the exact dominant eigenvalue and  $\lambda^{(k)}$  is the  $k$ th approximation, the **relative error** in  $\lambda^{(k)}$  is  $|\lambda - \lambda^{(k)}|/|\lambda|$ . In practice,  $\lambda$  is unknown, so we use the **estimated relative error**:

$$\left| \frac{\lambda^{(k)} - \lambda^{(k-1)}}{\lambda^{(k)}} \right| < E \quad (13)$$

to stop the iteration when accuracy  $E$  is achieved.

### EXAMPLE 4 | Estimated Relative Error

For the computations in Example 3, find the smallest  $k$  for which the estimated percentage error in  $\lambda^{(k)}$  is less than 0.1%.

**SOLUTION:**

| $\lambda^{(k)}$                 | $\frac{\lambda^{(k)} - \lambda^{(k-1)}}{\lambda^{(k)}}$ |                   | Estimated % error |
|---------------------------------|---------------------------------------------------------|-------------------|-------------------|
| $\lambda^{(2)} \approx 4.99361$ | $\frac{4.99361 - 4.84615}{4.99361}$                     | $\approx 0.02953$ | 2.953%            |
| $\lambda^{(3)} \approx 4.99974$ | $\approx 0.00123$                                       |                   | 0.123%            |
| $\lambda^{(4)} \approx 4.99999$ | $\approx 0.00005$                                       |                   | 0.005%            |
| $\lambda^{(5)} \approx 5.00000$ | $\approx 0.00000$                                       |                   | 0.000%            |

Thus  $\lambda^{(4)} = 4.99999$  is the first approximation with estimated percentage error below 0.1%.

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### Exercise Set 9.2

In Exercises 1–2, distinct eigenvalues are given. Determine whether  $A$  has a dominant eigenvalue, and if so, find it.

1. **a.**  $\lambda_1 = 7, \lambda_2 = 3, \lambda_3 = -8, \lambda_4 = 1$     **b.**  $\lambda_1 = -5, \lambda_2 = 3, \lambda_3 = 2, \lambda_4 = 5$

2. **a.**  $\lambda_1 = 1, \lambda_2 = 0, \lambda_3 = -3, \lambda_4 = 2$     **b.**  $\lambda_1 = -3, \lambda_2 = -2, \lambda_3 = -1, \lambda_4 = 3$

In Exercises 3–4, apply the power method with Euclidean scaling to  $A$ , starting with  $\mathbf{x}_0$ , stopping at  $\mathbf{x}_4$ .

3.  $A = \begin{bmatrix} 5 & -1 \\ -1 & -1 \end{bmatrix}; \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

4.  $A = \begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}; \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

In Exercises 5–6, apply the power method with maximum entry scaling to  $A$ , starting with  $\mathbf{x}_0$ , stopping at  $\mathbf{x}_4$ .

5.  $A = \begin{bmatrix} 1 & -3 \\ -3 & 5 \end{bmatrix}; \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

6.  $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 4 \end{bmatrix}; \quad \mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

7. Let  $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ ,  $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ .

- Apply the power method with maximum entry scaling; round to 3 decimal places; stop after 3 iterations.
- Use the result in (a) and the Rayleigh quotient to approximate the dominant eigenvalue.
- Find the exact dominant eigenvector and eigenvalue.
- Find the percentage error in the eigenvalue approximation from (b).

In Exercises 9–10, a matrix  $A$  with a dominant eigenvalue and a power sequence  $\mathbf{x}_0, A\mathbf{x}_0, \dots, A^5\mathbf{x}_0$  are given. Use Formulas (9) and (10) to approximate the dominant eigenvalue and a corresponding eigenvector.

9.  $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ ;  $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ ,  $A\mathbf{x}_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ ,  $A^2\mathbf{x}_0 = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$ ,  $A^3\mathbf{x}_0 = \begin{bmatrix} 13 \\ 14 \end{bmatrix}$ ,  $A^4\mathbf{x}_0 = \begin{bmatrix} 41 \\ 40 \end{bmatrix}$ ,  $A^5\mathbf{x}_0 = \begin{bmatrix} 121 \\ 122 \end{bmatrix}$

11. Consider  $A = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$  and  $\mathbf{x}_0 = \begin{bmatrix} a \\ b \end{bmatrix}$  (a unit vector,  $a \neq 0$ ). Show that even though  $A$  is symmetric and has a dominant eigenvalue, the power sequence (1) in Theorem 9.2.1 does not converge. This shows that the requirement that the dominant eigenvalue be positive is essential.

12. Use the power method with Euclidean scaling to approximate the dominant eigenvalue and a corresponding eigenvector of  $A$ ; stop when the estimated percentage error in the eigenvalue approximation is less than 0.1%.

a.  $A = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 4 & -1 \\ 3 & -1 & 10 \end{bmatrix}$

15. Prove: If  $A$  is a nonzero  $n \times n$  matrix, then  $A^T A$  and  $A A^T$  have positive dominant eigenvalues.